Problem 1

P2

$$\varepsilon(\omega) = \varepsilon_0\left(1+\chi_e(\omega)\right), \quad \varepsilon'' = \varepsilon_0\,\chi_e'' = \frac{\beta_0\omega\tau}{1+\omega^2\tau^2}.$$

Kramers-Kronig $\quad \chi'(\omega) = \frac{2}{\pi} P\int_0^\infty \frac{x\,\chi''(x)}{x^2-\omega^2}\,dx$

$$\chi'(\omega) = \frac{2}{\pi}\,\frac{\beta_0}{\varepsilon_0}\,P\int_0^\infty dx\; x\,\frac{x\tau}{1+x^2\tau^2}\,\frac{1}{x^2-\omega^2} = \begin{pmatrix} y = x\tau \\ dx = \frac{dy}{\tau}\end{pmatrix}$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty \frac{dy}{\tau}\,\frac{y^2}{\tau(1+y^2)}\,\frac{\tau^2}{y^2-\omega^2\tau^2} =$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty dy\,\frac{y^2}{1+y^2}\,\frac{1}{y^2-\omega^2\tau^2} = \left(\text{denote } \omega^2\tau^2 = u^2\right)$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty dy\left(\frac{1}{1+y^2}+\frac{u^2}{y^2-u^2}\right)\frac{1}{1+u^2}$$

$$\frac{y^2}{(1+y^2)(y^2-u^2)} = \left(\frac{1}{1+y^2}+\frac{u^2}{y^2-u^2}\right)\frac{1}{1+u^2}.$$

$$I_1 = P\int_0^\infty dy\,\frac{1}{1+y^2} = \frac{\pi}{2}$$ (Principal part "P" does not play a role since the integral is regular)

$$I_2 = P\int_0^\infty \frac{dy}{y^2-u^2} = (y=ux) = \frac{1}{u}\,P\int_0^\infty dx\,\frac{1}{x^2-1} = \frac{1}{u}\int_0^\infty dx\,\frac{1}{2}\left(\frac{1}{x-1}-\frac{1}{x+1}\right)$$

$$= \frac{1}{2u}\left\{\int_0^{1-\varepsilon}\frac{dx}{x-1} + \int_{1+\varepsilon}^{L}\frac{dx}{x-1} - \int_0^{L}\frac{dx}{x+1}\right\}.$$

At the end we need to take limits: $\varepsilon\to+0$, $L\to\infty$

$$\int_0^{1-\varepsilon}\frac{dx}{x-1} = (x=1-z) = \int_1^{\varepsilon}\frac{-dz}{-z} = -\int_\varepsilon^1 d(\ln z) = \ln(\varepsilon)$$

$$\int_{1+\varepsilon}^{L} \frac{dx}{x-1} = (x = 1+z) = \int_{\varepsilon}^{L-1} \frac{dz}{z} = \ln \frac{L-1}{\varepsilon}.$$

$$\int_{0}^{L} \frac{dx}{x+1} = \ln\left(\frac{L+1}{1}\right) = \ln(L+1)$$

$$\Rightarrow I_2 = \frac{1}{2u}\left\{\ln(\varepsilon) + \ln\left(\frac{L-1}{\varepsilon}\right) - \ln(L+1)\right\} = (\varepsilon \text{ drops out!})$$

$$= \frac{1}{2u} \ln\left(\frac{L-1}{L+1}\right) \xrightarrow[L\to\infty]{} 0.$$

Thus $\chi'(\omega) = \frac{2\beta_0}{\pi\varepsilon_0} \frac{\pi}{2} \frac{1}{1+(\omega\tau)^2} = \frac{\beta_0}{\varepsilon_0(1+\omega^2\tau^2)}$

$$\chi(\omega) = \chi' + i\chi'' = \frac{\beta_0}{\varepsilon_0(1+\omega^2\tau^2)}(1+i\omega\tau) =$$

$$= \frac{\beta_0(1+i\omega\tau)}{\varepsilon_0(1+i\omega\tau)(1-i\omega\tau)} = \frac{\beta_0}{\varepsilon_0(1-i\omega\tau)}$$

$$\Rightarrow \varepsilon(\omega) = \varepsilon_0 \chi(\omega) = \frac{\beta_0}{1-i\omega\tau}.$$

Problem 2

$$E_x = A e^{ikx-i\omega t} \begin{cases} \exp\left(-z\sqrt{k^2-\omega^2/c^2}\right), & z>0 \\ \exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right), & z<0 \end{cases} \qquad (1)$$

Similarly, E_z is given in the class notes:

$$E_z = ikAe^{ikx-i\omega t} \begin{cases} \dfrac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}, & z>0 \\ -\dfrac{\exp\left(z\sqrt{k^2-\varepsilon\omega^2/\varepsilon_0 c^2}\right)}{\sqrt{k^2-\varepsilon\omega^2/\varepsilon_0 c^2}}, & z<0 \end{cases} \qquad (2)$$

Magnetic field is given by

$$\vec{B} = -\frac{i}{\omega}\nabla\times\vec{E}$$

$$B_x = -\frac{i}{\omega}\left(\frac{\partial}{\partial y}E_z - \frac{\partial E_y}{\partial z}\right) = 0$$

(since $E_y=0$ and E_x, E_z are independent of y)

$$B_z = -\frac{i}{\omega}\left(\frac{\partial}{\partial x}E_y - \frac{\partial E_x}{\partial y}\right) = 0$$

$$B_y = -\frac{i}{\omega}\left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) = -\frac{i}{\omega}\left(\frac{\partial E_x}{\partial z} - ikE_z\right) \qquad (3)$$

We now substitute Eqs. (1) and (2) into Eq. (3):

for $z>0$:

$$B_y = -\frac{i}{\omega}\left(-\sqrt{k^2-\omega^2/c^2}\,Ae^{ikx-i\omega t}\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right) + k^2Ae^{ikx-i\omega t}\frac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}\right) = \qquad (4)$$

$$= -\frac{i\omega}{c^2} A e^{ikx-i\omega t} \frac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}$$

for $z<0$:

$$B_y = -\frac{i}{\omega}\left(\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\, A e^{ikx-i\omega t} \exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)\right.$$

$$\left. - k^2 A e^{ikx-i\omega t} \frac{\exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)}{\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}}\right) \qquad (5)$$

$$= \frac{i\varepsilon\omega}{\varepsilon_0 c^2} A e^{ikx-i\omega t} \frac{\exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)}{\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}}$$

By virtue of the dispersion equation for the surface plasmons:

$$\frac{\varepsilon}{\varepsilon_0} = -\frac{\sqrt{k^2-\frac{\omega^2\varepsilon}{c^2\varepsilon_0}}}{\sqrt{k^2-\frac{\omega^2}{c^2}}}$$

the two expressions (4) and (5) match at $z=0$.

Thus, $\vec{B}$ is indeed continuous at the surface.

Problem 3

$$E_y = E_y(z)\, e^{ikx - i\omega t}$$

Maxwell equation (3) of the notes for E_y is completely analogous to the one for E_x with the same boundary condition (continuity at $z=0$).

Thus

$$E_y = A e^{ikx - i\omega t} \begin{cases} \exp\left(-z\sqrt{k^2 - \frac{\omega^2}{c^2}}\right), & z>0 \\ \exp\left(z\sqrt{k^2 - \frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right), & z<0 \end{cases}$$

Since $E_x = 0$ and $E_z = 0$ ($D_z = 0$) <u>all</u> relevant components of $\vec{E}$ and $\vec{D}$ are properly continuous at $z=0$. Let us check the continuity of $\vec{B}$ though:

$$B_x = -\frac{i}{\omega}(\nabla \times \vec{E})_x = -\frac{i}{\omega}\left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right) = \frac{i}{\omega}\frac{\partial E_y}{\partial z}$$

$$B_x = \frac{iA}{\omega} e^{ikx - i\omega t} \begin{cases} -\sqrt{k^2 - \frac{\omega^2}{c^2}}\, \exp\left(-z\sqrt{k^2 - \frac{\omega^2}{c^2}}\right), & z>0 \\ \sqrt{k^2 - \frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\, \exp\left(z\sqrt{k^2 - \frac{\omega^2}{c^2}}\right), & z<0 \end{cases}$$

This would require that

$$\sqrt{k^2 - \frac{\omega^2}{c^2}} = -\sqrt{k^2 - \frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}$$

Such a solution is impossible (at least when $\sqrt{k^2 - \omega^2/c^2}$ and $\sqrt{k^2 - \frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}$ are real). $\longrightarrow$ no surface TE-wave.

$$\sigma(\omega) = \frac{ie^2 n}{m\left(\omega + \frac{i}{\tau}\right)}$$

Fourier transform applied electric field

$$\vec{E}(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t}\, \vec{E}(t) = \int_{0}^{\infty} dt\, e^{i\omega t}\, \vec{E}_0\, \theta(t) =$$

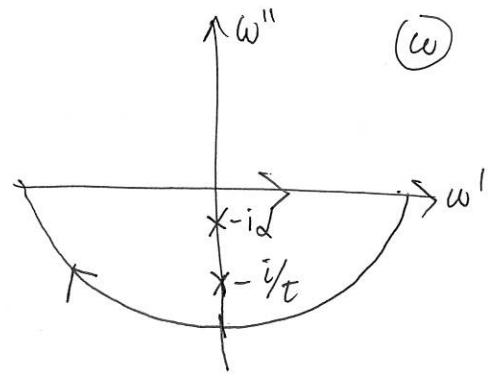
$$= \vec{E}_0 \int_0^{\infty} dt\, e^{i\omega t - \alpha t} = \vec{E}_0 \left. \frac{e^{-t(\alpha - i\omega)}}{-(\alpha - i\omega)} \right|_0^{\infty} = \frac{\vec{E}_0}{\alpha - i\omega}$$

which is saying that all frequencies are present.

By Ohm's law $\vec{j}(\omega) = \sigma(\omega)\, \vec{E}(\omega) = \dfrac{ie^2 n}{m\left(\omega + \frac{i}{\tau}\right)} \dfrac{i\vec{E}_0}{\omega + i\alpha}$

Now transform this back to time domain:

$$\vec{j}(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi}\, e^{-i\omega t}\, \vec{j}(\omega) = -\frac{e^2 n \vec{E}_0}{m} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{\left(\omega + \frac{i}{\tau}\right)(\omega + i\alpha)}$$



$t > 0 \Rightarrow$ need $e^{-i\omega t}$ to converge:

$e^{-it(\omega' + i\omega'')} = e^{-i\omega' t + \omega'' t}$

$\Rightarrow$ can close the contour in $\omega'' < 0$ half-plane

$$\vec{j}(t) = -\frac{e^2 n \vec{E}_0}{m} \frac{(-2\pi i)}{2\pi} \sum_{\text{Res}} \frac{e^{-i\omega t}}{\left(\omega + \frac{i}{\tau}\right)(\omega + i\alpha)} = \frac{i e^2 n \vec{E}_0}{m} \cdot$$

$$\cdot \left\{ \frac{e^{-t/\tau}}{i\alpha - \frac{i}{\tau}} + \frac{e^{-\alpha t}}{\frac{i}{\tau} - i\alpha} \right\} = \frac{e^2 n \vec{E}_0}{m\left(\frac{1}{\tau} - \alpha\right)} \left(e^{-\alpha t} - e^{-t/\tau}\right) =$$ can take the limit $\alpha \to 0$ now

$$= \frac{e^2 n \tau}{m} \vec{E}_0 \left(1 - e^{-t/\tau}\right) = \left(1 - e^{-t/\tau}\right) \sigma_0 \vec{E}_0 \; ; \; \sigma_0 = \frac{e^2 n \tau}{m}.$$

$\vec{j}(t) \to \sigma_0 \vec{E}_0$ for $t \gg \tau$.

P5

Account of the finite size of conductor, and of the induced field due to the accumulated charges leads to $\vec{j}(\omega) = \frac{i n e^2}{m} \frac{\omega}{(\omega^2-\omega_p^2)+i\omega\gamma} \vec{E}_0(\omega)$ ← applied field

(see March 1 discussion note). $(\gamma \equiv \frac{1}{\tau})$.

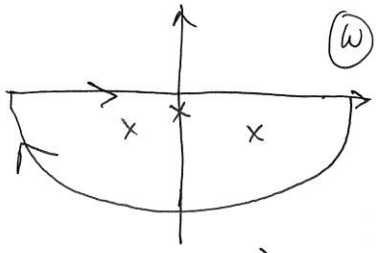
Since $\vec{E}_0(\omega) = \frac{i\vec{E}_0}{\omega+i\alpha}$, we now need

$$\vec{j}(t) = -\frac{ne^2\vec{E}_0}{m}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\frac{e^{-i\omega t}\,\omega}{[(\omega^2-\omega_p^2)+i\omega\gamma][\omega+i\alpha]}$$

Poles are at $\omega^2+i\omega\gamma-\omega_p^2=0$

$$\omega_{1,2} = \frac{-i\gamma \pm \sqrt{(i\gamma)^2+4\omega_p^2}}{2} = -\frac{i}{2}\gamma \pm \sqrt{\omega_p^2-\frac{\gamma^2}{4}}$$

and at $\omega=-i\alpha$ $\qquad = -\frac{i}{2}\gamma \pm \Omega$.



$$\vec{j}(t) = -\frac{e^2 n\vec{E}_0}{m}\left(\frac{-2\pi i}{2\pi}\right)\sum_{\text{Res}}\frac{\omega e^{-i\omega t}}{(\omega-\omega_1)(\omega-\omega_2)(\omega+i\alpha)} =$$

$$= \frac{ie^2n\vec{E}_0}{m}\left\{\frac{\omega_1 e^{-i\omega_1 t}}{(\omega_1-\omega_2)(\omega_1+i\alpha)} + \frac{\omega_2 e^{-i\omega_2 t}}{(\omega_2-\omega_1)(\omega_2+i\alpha)} + \frac{-i\alpha e^{-\alpha t}}{(-i\alpha-\omega_1)(-i\alpha-\omega_2)}\right\}$$

Set $\alpha=0 \Rightarrow \vec{j}(t) = \frac{ie^2n\vec{E}_0}{m}\frac{e^{-i\omega_1 t}-e^{-i\omega_2 t}}{\omega_1-\omega_2} =$

$$= \frac{ie^2n\vec{E}_0}{m}\frac{e^{-\gamma t/2}\left(e^{-i\Omega t}-e^{i\Omega t}\right)}{-\frac{i}{2}\gamma+\Omega-(-\frac{i}{2}\gamma-\Omega)} = \frac{e^2n\vec{E}_0}{m}\frac{\sin(\Omega t)\,e^{-\gamma t/2}}{\Omega} =$$

$$= \frac{\sin(\Omega t)}{\Omega\tau}e^{-\frac{t}{2\tau}}\sigma_0\vec{E}_0.$$

Observe that $\Omega = \sqrt{\omega_p^2 - \frac{\gamma^2}{4}} = \frac{\gamma}{2}\sqrt{4\omega_p^2\tau^2 - 1}$

$$\Omega = \frac{\sqrt{(2\omega_p\tau)^2 - 1}}{2\tau} = \frac{\sqrt{(\omega_p\tau)^2 - \frac{1}{4}}}{\tau}$$

$$\vec{j}(t) = \frac{\sin\left[\sqrt{(\omega_p\tau)^2 - \frac{1}{4}}\,\frac{t}{\tau}\right] e^{-\frac{t}{2\tau}}}{\sqrt{(\omega_p\tau)^2 - \frac{1}{4}}}\,\sigma_0 \vec{E}_0 .$$

(1) First check the limit $\omega_p \to 0$: $\sqrt{(\omega_p\tau)^2 - \frac{1}{4}} \to \sqrt{-\frac{1}{4}} = \frac{i}{2}$

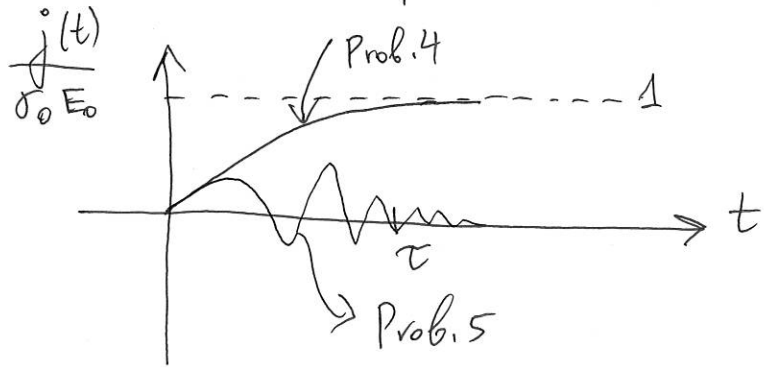
$$\sin\left(\frac{it}{2\tau}\right) = \frac{e^{i\left(\frac{it}{2\tau}\right)} - e^{-i\left(\frac{it}{2\tau}\right)}}{2i} = \frac{e^{\frac{t}{2\tau}} - e^{-\frac{t}{2\tau}}}{-2i}$$

$$\vec{j}(t) = \frac{e^{\frac{t}{2\tau}} - e^{-\frac{t}{2\tau}}}{-2i} \cdot \left(\frac{2}{i}\right) \cdot e^{-\frac{t}{2\tau}}\,\sigma_0 \vec{E}_0 =$$

$$= \left(1 - e^{-t/\tau}\right)\sigma_0 \vec{E}_0$$ (which is exactly the result of Prob. 4)

(2) Let now $\omega_p\tau \gg 1 \Rightarrow \sqrt{(\omega_p\tau)^2 - \frac{1}{4}} \to \omega_p\tau$.

$$\vec{j}(t) = \frac{\sin(\omega_p t)\, e^{-t/2\tau}}{\omega_p\tau}\,\sigma_0 \vec{E}_0 \longrightarrow 0$$ for $t \gg \tau$.



Account of the induced field leads to the disappearance of the current, as it should be, since <u>local</u> electric field inside the conductor is <u>zero</u>.